Functional Test Plan for Arguably

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Revision History

Name	Date	Reason For Change	Version
Jason, Aadhav, Praveen, WenHao	3/29/26	Initial Draft	1.0 draft 1

1 Introduction

This document serves as an outline of the goals for testing of the Arguably debate platform. It will describe all the functional tests for the features within the application and how we plan on implementing them. Functional testing areas include user authentication, room creation and configuration, session-based role assignment, structured turn-taking and debate flow, audience queue management, real-time moderation controls, live transcription, AI-assisted claim detection, and debate replay with analytics. The objectives of this document are to outline tests and create detailed guidelines for the functional testing plan. The other objective is to confirm that each test case verifies the software and fulfills the requirements described in our SRS document.

2 Project and Scope

This document presents the test cases which our team will execute so that the Arguably application can be fully tested. It will contain descriptions of how each of the use cases and functional requirements will be tested. We will test Arguably using a combination of automated tests (Jest unit tests, Playwright end-to-end tests) and manual testing for real-time WebRTC and socket.io interactions. For specific tests, reference the Test Specifications and Functional Tests sections of this document. After completing the code for each feature, we will begin to test it. Once we have finished testing that feature, we will continue to implement another section of our project.

3 Test Design Specifications

In this document, we will have specific tests organized by feature area as defined in the SRS. Each test will be simulated with a user input and then go through steps to get the expected output of the program. If all the steps of the program produce the correct output for the input, then the test will count as being passed. Any test that fails somewhere along the way to the final step of testing will be considered as a failed test. Tests are designed to be executable by any team member or external tester with access to the deployed application and a supported browser.

3.1 Testing Environment and Tools

- **Browsers:** Latest two stable versions of Chrome, Firefox, Edge (Chromium), and Safari
- **Automated Unit Tests:** Jest with TypeScript for server actions, utility functions, and Prisma queries
- **End-to-End Tests:** Playwright for browser-based UI flows
- **Real-Time Testing:** Manual testing with multiple browser sessions for WebRTC, socket.io events, and debate flow

- **Database:** PostgreSQL via Supabase (test environment with seeded data)
- **OS:** macOS / Linux / Windows (cross-platform via browser)
- **Node.js:** Version 20+

3.2 Assumptions and Special Considerations

- Users have access to the Internet, a supported web browser, camera, and microphone.
- At least three non-admin user accounts are active in the test environment.
- Supabase authentication services are available during testing.
- The real-time server (mediasoup SFU + socket.io) is running and accessible.
- WebRTC tests require two or more browser sessions (or devices) to validate peer-to-peer media flows.
- AI features (Live Transcription, Claim Detection) depend on external API availability (OpenAI Whisper, Gemini LLM); tests will verify graceful degradation if unavailable.

4 Test Specifications

4.1 User Registration and Login

1. The user navigates to the `/auth` page.
2. The application displays a login/signup form with fields for email and password.
3. The user fills out the form and submits.
   - If the user does not fill out a required field or provides an invalid email format, the application prevents submission and displays a validation error.
4. For signup: the application creates a Supabase Auth account, upserts a User record in the database via `ensureUserProfile`, and redirects the user to the browse page.
5. For login: the application validates credentials against Supabase Auth, sets an HTTP-only JWT cookie, and redirects the user to the browse page.

4.2 Room Creation and Configuration

1. The authenticated user navigates to the `/create` page via the main navigation.
2. The application displays a multi-step room creation form with fields for room name, description, debate format, turn duration, team size (if Team format), and audience elimination voting settings.
3. The user completes the form and submits.
   - If the user selects Team format, the team size field (2–5) is displayed and required.
   - If the user enables audience elimination voting, the vote threshold percentage field is displayed.
4. The application creates a Session record in the database, generates a unique join code, assigns the creator as Host, and redirects to the room page at `/room/[code]`.
5. The join code is displayed to the Host for sharing purposes.

4.3 Browse and Join Rooms

1. The authenticated user navigates to the `/browse` page.
2. The application displays a list of available debate rooms with filters for format, status, and search.
3. The user selects a room and clicks join, or enters a join code directly.
4. The application creates a ParticipatesIn record, assigns the appropriate role (Debater if slots available, otherwise Audience), and redirects to the room page.

4.4 Session-Based Role Assignment

1. A user creates a room and is automatically assigned the Host/Creator role.
2. The Host selects a participant and assigns them the Moderator role.
3. The application updates the ParticipatesIn record and the Session's moderatorId.
4. The assigned user's UI immediately updates to show Moderator controls without page refresh.
5. When new users join, they are assigned Debater (if slots available) or Audience role automatically.
6. All role transitions are logged with timestamps in the ParticipatesIn table.

4.5 Structured Turn-Taking and Debate Flow

1. The Moderator or Host clicks "Start Debate" in the room.
2. The server-side debate engine initializes turn order based on the selected format.
3. A countdown timer is displayed to all participants showing minutes and seconds remaining.
4. When the timer expires, the current speaker is automatically muted, a 3-second grace period occurs, and the next speaker is unmuted.
5. The Moderator can pause the debate (stopping all timers), resume it (with a 3-second countdown), or advance the turn manually.
6. Visual warnings appear at 30 seconds remaining (color change) and audio warnings at 10 seconds.
7. The debate can be ended by the Moderator, at which point all timers stop and the session status is updated.

4.6 Audience Queue and Dynamic Replacement

1. An audience member clicks "Join Queue" to enter the speaker queue.
2. The application adds the user to the FIFO queue and displays their position.
3. When a debater slot opens (e.g., turn ends in Expert vs. Crowd), the system automatically promotes the next queued audience member within 5 seconds.
4. The promoted user receives a 10-second window to accept; if they decline or the window expires, they are moved to the back of the queue and the next user is promoted.
5. If audience kick voting is enabled, audience members can vote to remove a speaker; if the configured threshold is reached, the speaker is removed and replaced from the queue.

4.7 Real-Time Moderation Controls

1. The Moderator sees a control panel with pause, resume, mute, kick, skip turn, extend turn, and promote controls.
2. The Moderator clicks "Pause" and all debate timers immediately stop; a "Debate Paused" indicator is shown to all participants.
3. The Moderator clicks "Resume" and a 3-second countdown appears before timers restart.
4. The Moderator mutes a participant's microphone; the muted user cannot unmute themselves until the Moderator grants permission.
5. The Moderator removes a participant from the room, optionally specifying a reason and ban.
6. The Moderator extends the current turn in 30-second increments (up to 5 minutes maximum extension).
7. The Moderator manually promotes a specific audience member to replace a debater.
8. All moderation actions are logged with timestamps.

4.8 Live Transcription

1. A debate is in progress with active speakers.
2. The real-time server streams the active speaker's audio to OpenAI Whisper.
3. Transcribed text appears in the transcript panel within 3 seconds of speech, attributed to the correct speaker with their name and a timestamp in MM:SS format.
4. The transcript panel auto-scrolls to show the latest content; users can scroll back to view previous content.
5. Users can collapse or expand the transcript panel.
6. The Moderator can correct speaker misattributions in the transcript.

4.9 AI-Assisted Claim Detection and Source Surfacing

1. The transcript stream is analyzed by the Gemini LLM for factual claims (statistics, studies, laws, verifiable facts).
2. Detected claims are highlighted in the transcript within 10 seconds of the statement being transcribed.
3. Relevant sources from reputable domains (news organizations, academic institutions, government websites) are retrieved and displayed within 15 seconds of claim detection.

4. Sources are displayed as contextual links near the relevant claim, including the source title, URL, and a brief snippet, labeled "AI-suggested."

5. Users can dismiss or report irrelevant sources.

6. An optional claim accuracy indicator shows "Supported," "Disputed," or "Unverified" based on source analysis.

4.10 Debate Replay and Participation Analytics

1. After a debate ends, the system stores the recording (video, audio, transcript data) for a minimum of 30 days.

2. Users navigate to the replay interface, which provides synchronized video and transcript playback.

3. Users can click on transcript segments to jump to the corresponding point in the video.

4. Playback controls include play, pause, seek, and adjustable playback speed.

5. Role changes, eliminations, and significant events are marked on the replay timeline.

6. The system displays per-participant analytics: total speaking time (absolute and percentage), number of turns, and average turn duration.

7. Users can export turn and timing data as CSV.

8. Each user's participation history shows past debates with date, topic, role, and outcome, and can be set to public or private.

5 Functional Tests

5.1 Authentication Tests

Test	Description	Setup / Preconditions	Expected Result
AUTH-1	Register with valid email and password	Navigate to `/auth` page; enter a valid unused email and a password meeting minimum requirements	Account is created in Supabase Auth; User record is upserted in DB; user is redirected to `/browse` with active session
AUTH-2	Register with an already-used email	Navigate to `/auth` page; enter an email that is already registered	Application displays an error message indicating the email is already in use; no duplicate account is created
AUTH-3	Login with valid credentials	Navigate to `/auth` page; enter valid email and password for an existing account	User is authenticated; JWT cookie is set; user is redirected to `/browse`
AUTH-4	Login with invalid credentials	Navigate to `/auth` page; enter a valid email with an incorrect password	Application displays an authentication error; no session is created
AUTH-5	Access protected route without login	Ensure no active session; navigate directly to `/create` or `/room/[code]`	Middleware redirects user to `/auth` page
AUTH-6	Login with empty fields	Navigate to `/auth` page; submit the form with email or password fields empty	Form validation prevents submission; error message displayed for missing fields

5.2 Room Creation and Configuration Tests

Test	Description	Setup / Preconditions	Expected Result
RC-1	Create room with valid settings (REQ-1.1, REQ-1.2)	Authenticated user navigates to `/create`; fills in room name, selects One-on-One format, default turn duration	Session record is created in DB; unique join code is generated and displayed; user is redirected to `/room/[code]`
RC-2	Host role auto-assigned on creation (REQ-1.3)	User creates a new room	ParticipatesIn record is created with `sessionRole = CREATOR`; Session `hostId` is set to the creating user
RC-3	All four debate formats available (REQ-1.4)	Navigate to `/create` page	Format selection shows Expert vs. Crowd, One-on-One, Team, and Panel options
RC-4	Configure turn duration (REQ-1.5)	Navigate to `/create`; adjust turn duration slider/input	Turn duration can be set between 30 seconds and 30 minutes; default is 2 minutes; value is saved to Session record
RC-5	Configure team size for Team format (REQ-1.6)	Select Team format on `/create` page	Team size field appears allowing values between 2 and 5; value is saved to Session record
RC-6	Configure audience elimination voting (REQ-1.7)	Navigate to `/create`; toggle audience elimination voting on	Vote threshold percentage field appears; value is configurable and saved to Session record
RC-7	Format locked after debate begins (REQ-1.8)	Create a room with One-on-One format; start the debate via Moderator controls	Attempting to change the debate format after the debate has started is prevented; format field is disabled or hidden

Test	Description	Setup / Preconditions	Expected Result
RC-8	Room appears in browse listing (REQ-1.9)	Create a new room	Within 5 seconds, the room appears in the `/browse` page listing for other authenticated users
RC-9	Create room without required fields	Navigate to `/create`; submit form without entering a room name	Form validation prevents submission; error message indicates required fields

5.3 Session-Based Role Assignment Tests

Test	Description	Setup / Preconditions	Expected Result
RA-1	Host auto-assigned on room creation (REQ-2.1)	User creates a new debate room	User's ParticipatesIn record has `sessionRole = CREATOR`; role persists for the duration of the session
RA-2	Host assigns Moderator (REQ-2.2)	Host is in a room with at least one other participant	Host selects a participant and assigns Moderator role; Session `moderatorId` is updated; participant's ParticipatesIn record is updated to `MODERATOR`
RA-3	Role change notification (REQ-2.3)	Host assigns Moderator role to a participant	The assigned participant sees a notification of their role change and their UI updates to show Moderator controls within 2 seconds
RA-4	Debater role auto-assigned (REQ-2.4)	A user joins a room that has available debater slots	User is assigned the Debater role in ParticipatesIn; user has access to audio/video controls and speaking queue

Test	Description	Setup / Preconditions	Expected Result
RA-5	Audience role when debater slots full (REQ-2.5)	A user joins a room where all debater positions are filled	User is assigned the Audience role; audio and video capabilities are restricted until promoted
RA-6	Only one Host and one Moderator per room (REQ-2.6)	A room has an existing Host and Moderator	Attempting to assign a second Host or Moderator replaces the existing assignment rather than creating a duplicate; only one of each exists at any time
RA-7	Immediate permission update on role change (REQ-2.7)	A participant's role is changed from Audience to Debater	The participant's UI updates immediately to reflect Debater controls (audio/video) without requiring a page refresh
RA-8	Role transition logging (REQ-2.8)	Any role change occurs in a session	The ParticipatesIn record is updated with a timestamp; the initiating action is logged for audit purposes

5.4 Structured Turn-Taking and Debate Flow Tests

Test	Description	Setup / Preconditions	Expected Result
TT-1	Expert vs. Crowd format (REQ-3.1)	Create a room with Expert vs. Crowd format; start debate with an expert and queued audience members	Expert maintains speaking position; challengers rotate from the audience FIFO queue after each turn
TT-2	One-on-One format (REQ-3.2)	Create a room with One-on-One format; two debaters join; Moderator starts debate	Exactly two debaters alternate speaking turns under moderator control

Test	Description	Setup / Preconditions	Expected Result
TT-3	Team format (REQ-3.3)	Create a room with Team format (team size 3); two teams join; start debate	Two teams of 3 alternate speaking turns; time allocation is shared within each team
TT-4	Panel format (REQ-3.4)	Create a room with Panel format; 4 panelists join; start debate	4 panelists speak in circular order with equal time allocation per turn
TT-5	Countdown timer display (REQ-3.5)	A debate is in progress	All connected participants see a countdown timer displaying minutes and seconds remaining for the current turn
TT-6	Visual and audio warnings (REQ-3.6)	A turn is in progress; timer counts down to 30 seconds and then 10 seconds	At 30 seconds remaining, the timer changes color (visual warning); at 10 seconds remaining, an audio warning is played
TT-7	Auto-mute on turn expiry (REQ-3.7)	A speaker's turn timer reaches zero	Within 1 second, the current speaker is automatically muted and the next speaker is unmuted
TT-8	Grace period between turns (REQ-3.8)	A turn expires and the next turn begins	A 3-second grace period occurs between turns before the next speaker's timer starts
TT-9	Turn state synchronization (REQ-3.9)	A debate is in progress with multiple connected participants	All participants see the same current speaker and timer value; a newly connected participant sees the correct current state
TT-10	Turn transition logging (REQ-3.10)	A turn advances (automatically or manually)	The turn transition is logged with a timestamp in the system for post-debate review

5.5 Audience Queue and Dynamic Replacement Tests

Test	**Description**	**Setup / Preconditions**	**Expected Result**
AQ-1	FIFO queue ordering (REQ-4.1)	Three audience members join the queue in order: User A, User B, User C	Queue displays users in FIFO order: A first, B second, C third
AQ-2	Queue position display (REQ-4.2)	An audience member joins the queue	The user can see their current position in the queue (e.g., "Position 3 of 5")
AQ-3	Join/leave queue without leaving room (REQ-4.3)	An audience member is in a room	User can join the queue and later leave the queue while remaining in the room as an audience member
AQ-4	Queue position preserved on disconnect (REQ-4.4)	An audience member is in queue position 2; they temporarily lose connection for 15 seconds	Upon reconnection within 30 seconds, their queue position is preserved at position 2
AQ-5	Automatic promotion (REQ-4.5)	A debater slot opens (e.g., challenger's turn ends in Expert vs. Crowd format)	The next queued audience member is automatically promoted within 5 seconds
AQ-6	Promotion acceptance window (REQ-4.6)	A queued user is promoted to debater	The promoted user has a 10-second window to accept their promotion; UI shows a countdown for acceptance
AQ-7	Skipped user moved to back (REQ-4.7)	A promoted user does not accept within 10 seconds	The skipped user is moved to the back of the queue (not removed entirely); the next user in queue is promoted

Test	Description	Setup / Preconditions	Expected Result
AQ-8	Audience kick voting (REQ-4.8)	Audience elimination voting is enabled with 60% threshold; 10 audience members are present	Audience members can cast anonymous votes to kick the current speaker; when 6 out of 10 vote, the speaker is removed and replaced from queue
AQ-9	Cooldown after elimination (REQ-4.9)	A user is eliminated via audience kick vote; cooldown is set to 5 minutes	The eliminated user cannot rejoin the queue until 5 minutes have passed; attempting to rejoin before cooldown expires shows an error with remaining time

5.6 Real-Time Moderation Controls Tests

Test	Description	Setup / Preconditions	Expected Result
MC-1	Pause debate (REQ-5.1)	A debate is in progress; Moderator is present	Moderator clicks "Pause"; all debate timers immediately stop
MC-2	Paused indicator displayed (REQ-5.2)	Moderator pauses the debate	All participants see a "Debate Paused" indicator on their screen
MC-3	Resume with countdown (REQ-5.3)	A debate is paused	Moderator clicks "Resume"; a 3-second countdown is displayed to all participants before timers restart
MC-4	Mute participant (REQ-5.4)	A debate is in progress with active speakers	Moderator clicks the mute button next to a participant; that participant's microphone is immediately muted with a single click

Test	Description	Setup / Preconditions	Expected Result
MC-5	Muted user cannot self-unmute (REQ-5.5)	A participant has been muted by the Moderator	The muted participant's unmute button is disabled; they cannot unmute until the Moderator grants permission
MC-6	Remove participant (REQ-5.6)	A debate is in progress with multiple participants	Moderator clicks "Remove" on a participant; the participant is immediately removed from the room
MC-7	Remove with reason and ban (REQ-5.7)	Moderator removes a participant	Moderator can optionally specify a removal reason and ban the user from rejoining the room
MC-8	Auto-replace on removal (REQ-5.8)	A debater is removed by the Moderator; audience queue has waiting members	The system automatically promotes the next audience member from the queue to replace the removed debater
MC-9	Skip current turn (REQ-5.9)	A debate is in progress; a speaker is mid-turn	Moderator clicks "Skip Turn"; the current speaker's turn ends immediately and the next speaker begins
MC-10	Extend current turn (REQ-5.10)	A debate is in progress; a speaker is mid-turn	Moderator clicks "Extend"; the turn is extended by 30 seconds; extension can be repeated up to a maximum total extension of 5 minutes
MC-11	Manual promote from queue (REQ-5.11)	Audience queue has multiple waiting members; a debater slot is available	Moderator selects a specific audience member from the queue and promotes them to debater, bypassing FIFO order

Test	Description	Setup / Preconditions	Expected Result
MC-12	Moderation action logging (REQ-5.12)	Any moderation action is performed (pause, mute, remove, etc.)	The action is logged with a timestamp for accountability and post-session review

5.7 Live Transcription Tests

Test	Description	Setup / Preconditions	Expected Result
LT-1	Real-time transcription (REQ-6.1)	A debate is in progress; a speaker is speaking clearly in English	Speech-to-text transcription is generated in real time and displayed in the transcript panel
LT-2	Transcription latency (REQ-6.2)	A speaker says a sentence during an active debate	Transcribed text appears in the transcript panel within 3 seconds of speech
LT-3	Transcription accuracy (REQ-6.3)	A speaker delivers clear English speech	Transcription achieves a minimum accuracy of 85% for clear speech
LT-4	Speaker attribution (REQ-6.4, REQ-6.5)	Multiple speakers take turns in a debate	Each transcript segment is attributed to the correct speaker with their name displayed; attribution accuracy is at least 95%
LT-5	Timestamp display (REQ-6.6, REQ-6.7)	Transcription segments are generated during a debate	Each segment has a timestamp in MM:SS format relative to the debate start time, accurate to within 1 second
LT-6	Transcript panel auto-scroll (REQ-6.8)	Transcription is actively generating new segments	The transcript panel auto-scrolls to show the latest content as it arrives
LT-7	Scroll back in transcript (REQ-6.9)	Several minutes of transcript have been generated	User can scroll up through the transcript panel to view previous content

Test	Description	Setup / Preconditions	Expected Result
LT-8	Collapse/expand transcript panel (REQ-6.10)	The transcript panel is visible during a debate	User can collapse the transcript panel to hide it, and expand it again to show it
LT-9	Moderator corrects misattribution (REQ-6.11)	A transcript segment is attributed to the wrong speaker	The Moderator can select the misattributed segment and correct the speaker attribution

5.8 AI-Assisted Claim Detection and Source Surfacing Tests

Test	Description	Setup / Preconditions	Expected Result
CD-1	Factual claim detection (REQ-7.1)	A speaker makes a factual claim (e.g., "The US GDP grew by 3% in 2025") during a debate with live transcription active	The system identifies the statement as a factual claim containing a verifiable statistic
CD-2	Claim detection latency (REQ-7.2)	A factual claim is transcribed	The claim is detected and flagged within 10 seconds of the statement being transcribed
CD-3	Claim highlighting (REQ-7.3)	A claim is detected in the transcript	The detected claim is visually highlighted in the transcript panel to draw attention
CD-4	Source retrieval from reputable domains (REQ-7.4)	A factual claim has been detected	The system retrieves relevant sources from reputable domains including news organizations, academic institutions, and government websites
CD-5	Source display latency (REQ-7.5)	A claim has been detected	Retrieved sources are displayed within 15 seconds of claim detection

Test	Description	Setup / Preconditions	Expected Result
CD-6	Source display format (REQ-7.6)	Sources have been retrieved for a claim	Sources appear as contextual links near the relevant claim, showing the source title, URL, and a brief snippet
CD-7	AI-suggested label (REQ-7.7)	Sources are displayed for a detected claim	All surfaced sources are labeled "AI-suggested" to indicate automated retrieval
CD-8	Dismiss/report sources (REQ-7.8)	Sources are displayed for a claim	Users can dismiss an irrelevant source (removing it from their view) or report it
CD-9	False positive rate (REQ-7.9)	Multiple statements are made during a debate, including both factual claims and opinions	The system's false positive rate for claim detection remains below 20%
CD-10	Claim accuracy indicator (REQ-7.10)	A claim has been detected and sources retrieved	An optional accuracy indicator displays "Supported," "Disputed," or "Unverified" based on source analysis

5.9 Debate Replay and Participation Analytics Tests

Test	Description	Setup / Preconditions	Expected Result
DR-1	Automatic recording (REQ-8.1, REQ-8.2)	A debate begins and then ends	The system captures video, audio, and transcript data; recording starts when the debate begins and stops when it ends
DR-2	Recording retention (REQ-8.3)	A debate recording was created 25 days ago	The recording is still accessible and playable; recordings are stored for a minimum of 30 days

Test	Description	Setup / Preconditions	Expected Result
DR-3	Synchronized replay (REQ-8.4)	A completed debate with recording is available	The replay interface shows synchronized video and transcript playback; transcript highlights in sync with the video position
DR-4	Transcript-to-video navigation (REQ-8.5)	A replay is in progress with transcript visible	User clicks on a transcript segment; the video jumps to the corresponding point in the recording
DR-5	Playback controls (REQ-8.6)	A replay is in progress	User can play, pause, seek to specific points, and adjust playback speed (e.g., 0.5x, 1x, 1.5x, 2x)
DR-6	Timeline event markers (REQ-8.7)	A completed debate had role changes and eliminations	The replay timeline shows markers for role changes, eliminations, and other significant events; clicking a marker jumps to that point
DR-7	Speaking time analytics (REQ-8.8)	A completed debate had multiple speakers	Analytics show total speaking time per participant as both absolute time and percentage of total debate time
DR-8	Turn count analytics (REQ-8.9)	A completed debate had multiple turn transitions	Analytics show the number of turns taken by each participant
DR-9	Average turn duration (REQ-8.10)	A completed debate had multiple turns per participant	Analytics show the average turn duration for each participant
DR-10	CSV export (REQ-8.11)	Analytics are displayed for a completed debate	User clicks "Export CSV"; a CSV file is downloaded containing turn and timing data for all participants

Test	Description	Setup / Preconditions	Expected Result
DR-11	Participation history (REQ-8.12)	A user has participated in multiple debates	The user's profile shows a participation history with date, topic, role, and outcome for each past debate
DR-12	History privacy setting (REQ-8.13)	A user has a participation history	The user can set their participation history to public or private from their settings; private history is not visible to other users

5.10 WebRTC and Real-Time Communication Tests

Test	Description	Setup / Preconditions	Expected Result
RTC-1	Audio/video stream established	Two users join the same room as debaters using Chrome	Both users can see and hear each other via WebRTC through the mediasoup SFU; video is at least 720p
RTC-2	Cross-browser compatibility	User A joins on Chrome; User B joins on Firefox	Both users can establish WebRTC connections and see/hear each other
RTC-3	New participant sees existing streams	A debate is in progress with 2 active speakers; a third user joins as audience	The new participant can see and hear all active video/audio streams
RTC-4	Participant disconnect handling	A debater loses connection during an active debate	Other participants are notified of the disconnect; the disconnected user's stream is removed; the system handles the vacancy per queue rules

Test	Description	Setup / Preconditions	Expected Result
RTC-5	Socket.io event synchronization	A Moderator starts a debate	All connected participants receive the `turnUpdate` socket event and display the same debate state within 1 second

5.11 Browse and Navigation Tests

Test	Description	Setup / Preconditions	Expected Result
BN-1	Browse rooms with filters	Multiple rooms exist with different formats and statuses; user navigates to `/browse`	User can filter rooms by format (Expert vs. Crowd, One-on-One, Team, Panel), by status (waiting, in progress), and by search query
BN-2	Join room via join code	A room exists with join code "ABC123"; user enters the code on the browse page	User is added to the room and redirected to `/room/ABC123` with appropriate role assigned
BN-3	Join room via browse listing	User clicks "Join" on a room in the browse listing	User is added to the room with appropriate role and redirected to the room page
BN-4	Navigation adapts to auth state	User is logged in and views the floating navigation bar	Navigation shows authenticated options (create room, browse, settings); after logout, navigation shows only public options

6 Defect Responsibility and Resolution

1. **Issue Tracking**
   (a) We plan on using GitHub's built-in issue tracking within our repository to track issues as it is easily accessible to all of our teammates and is a powerful enough tool to explain issues that need to be resolved.
2. **Responsibilities**

System Part	**Developer**	**Responsibilities**
Authentication and User Management	Jason Li	Supabase Auth integration, middleware route protection, user profile management, and related functional tests.
Room Creation, Role Assignment, and Browse	WenHao Huang	Room creation flow, session configuration, role assignment logic, browse/join functionality, and related functional tests.
Debate Flow Engine and Turn-Taking	Jason Li	Server-side debate engine, turn management, timer synchronization, format-specific logic, and related functional tests.
Real-Time Server (WebRTC/SFU)	Praveen Sureshkumar	mediasoup SFU, socket.io signaling, WebRTC transport management, and related functional tests.
Moderation Controls and Audience Queue	Aadhav S. Baradwaj	Moderator UI controls, mute/kick/promote logic, audience queue management, kick voting, and related functional tests.
AI Services (Transcription, Claim Detection)	Praveen Sureshkumar	OpenAI Whisper integration, Gemini LLM claim detection, source surfacing, transcript panel UI, and related functional tests.
Replay and Analytics	Aadhav S. Baradwaj	Recording pipeline, replay interface, participation analytics, CSV export, and related functional tests.

7 Exit Criteria

1. All automated unit tests (Jest) have run and passed.
2. All Playwright end-to-end tests have run and passed across Chrome, Firefox, and Edge.

3. All manual WebRTC and real-time communication tests have been executed and verified with at least two concurrent browser sessions.
4. All functional test cases in this document have been executed with pass/fail results recorded.
5. No critical or high-severity defects remain unresolved.
6. AI service tests (transcription, claim detection) have been verified for both normal operation and graceful degradation when external APIs are unavailable.